Mathematics In India (IKS 4001)
ASSIGNMENT-1

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Q1. Role of the Indus Valley Civilisation in Early Mathematics

The Indus Valley Civilisation (IVC), flourishing approximately 3300–1300 BCE, laid significant foundations for early mathematical thinking, though no formal written mathematical texts have been discovered.

Key mathematical contributions of the IVC include:

- City Planning and Geometry: Cities like Harappa and Mohenjo-daro were built on precise grid patterns with streets meeting at right angles, indicating a practical understanding of geometry and spatial reasoning.
- Standardised Brick Ratios: Harappan bricks were manufactured in a consistent ratio of 4:2:1 (length:breadth:thickness). Fifteen different sizes followed this ratio — the optimal ratio for efficient bonding — demonstrating knowledge of proportion.
- Weights and Measures: The IVC developed one of the earliest standardised systems of weights and measures. Archaeological finds reveal a binary-decimal hybrid progression (1, 2, 4, 8, 16, 32, 64, then 160, 200, 320, 640…), implying use of both binary and decimal counting.
- Linear Measurement Scales: Rulers excavated at Mohenjo-daro and Lothal show precisely spaced divisions, with the 'Indus inch' measuring 33.53 mm. The Sumerian 'sushi' is exactly half an Indus inch, suggesting ancient trade-linked metrological standards.
- Geometry in Construction: Wedge-shaped bricks used for circular water wells indicate knowledge of trapezoidal geometry and circumference estimation ($\pi \approx 3.1$).
- Space-filling Patterns: Seals exhibit complex geometric motifs — including the Japanese fan tiling pattern derived from close-packing of circles — indicating a sophisticated appreciation of symmetry and area equivalence.

In summary, the IVC demonstrates that mathematical concepts like measurement, ratio, proportion, and geometry existed in applied form long before formal texts, being embedded in craft, trade, and urban practice.

Q2. Differences: Mathematics in the Indus Valley Civilisation vs. Vedic Period

Aspect	Indus Valley Civilisation (c. 2600–1900 BCE)	Vedic Period (c. 1500–500 BCE)
Nature	Practical and applied	Theoretical and abstract
Purpose	Trade, city construction, weights & measures	Rituals, astronomy, and philosophy
Number System	Binary-decimal hybrid inferred from weights; no written numerals found	Well-developed decimal system; names for very large numbers
Geometry	Practical geometry: grid planning, brick ratios, circular wells	Systematic altar geometry in Sulba Sutras (squares, circles, Pythagorean triples)
Mathematical Texts	None discovered; knowledge inferred from artefacts	Vedas, Brahmanas, Sulba Sutras
Zero & Algebra	No evidence	Early conceptual roots leading to later developments
Astronomy	Very limited evidence	Strong link: Jyotisa for time, calendar, ritual scheduling

Aspect	Indus Valley Civilisation (c. 2600–1900 BCE)	Vedic Period (c. 1500–500 BCE)
Transmission	Embedded in craft and trade practice	Oral tradition (sutras), teacher–student lineage

Q3. Contribution of Pāṇini to Indian Intellectual Thought

Pāṇini (c. 5th century BCE) is the author of the Ashtadhyayi ("Eight Chapters"), the world's first formal generative grammar, describing the rules of Classical Sanskrit. His contribution to Indian intellectual thought is profound and far-reaching.

Key Contributions:

- Formal Rule-Based System: The Ashtadhyayi consists of approximately 4,000 highly compressed sutras (rules) that describe Sanskrit morphology and syntax in a rigorous, unambiguous way — comparable to modern formal grammars and programming languages.
- Algorithmic Thinking: Pāṇini's rules operate algorithmically — a given input (word root + suffix) is transformed through an ordered sequence of rules to produce correct output. This logic parallels modern computational linguistics and formal automata theory.
- Metarules and Recursion: He introduced metarules (rules about rules), zero morphemes, and context-sensitive rules — sophisticated linguistic concepts that anticipate ideas formalised only in 20th-century mathematics.
- Influence on Mathematics: The precision and abstraction of Pāṇini's approach influenced the entire tradition of Indian science. Indian mathematics — especially its sutra-based, procedural style — mirrors the Paninian method of short, dense, rule-based formulations.
- Sanskrit and Scientific Discourse: By standardising Sanskrit grammar, Pāṇini made it the vehicle for all subsequent scientific, philosophical, and mathematical communication in India for over two millennia.

In short, Pāṇini's Ashtadhyayi established a model of formal, rule-based, abstract reasoning that became a template for Indian mathematics: concise rules (sutras), procedural operations, and rigorous logical structure over narrative proofs.

Q4. Meaning of 'Śulba', Four Sulbakāras, and Role of the Sulba Sūtras

Meaning of 'Śulba':

The word 'Śulba' (or Sulba) derives from the Sanskrit root 'sulb' (or 'śulb'), meaning 'to measure' or 'cord / rope'. It refers to the cord or rope used by Vedic priests to measure and lay out fire altars (yajnavedikas). The Sulba Sutras are therefore the 'rules of the cord'.

Four Major Sulbakāras and Their Contributions:

- Baudhayana (c. 800 BCE): Authored the oldest and most important Sulba Sutra. He stated what is now known as the Pythagorean theorem: 'The diagonal of a rectangle produces (an area) equal to the sum of those produced by its two sides.' He also gave methods for constructing squares equal in area to a given circle and vice versa, and provided the first known approximation $\sqrt{2} \approx 1 + 1/3 + 1/(3\times4) - 1/(3\times4\times34) \approx 1.4142156$.
- Apastamba (c. 600 BCE): Gave a more organised treatment of geometric constructions, including methods to construct a square equal in area to a given rectangle, and a circle equal to a square. He also listed Pythagorean triples (e.g., 3-4-5, 5-12-13, 8-15-17).
- Katyayana (c. 300 BCE): Generalised several results of Baudhayana and Apastamba, stating the Pythagorean result more explicitly for isosceles right triangles and providing additional altar constructions including those of the Great Altar (Mahavedi).
- Manava (c. 750 BCE): Contributed methods for constructing fire altars of specific shapes (falcon-shaped, etc.) and gave approximations for square roots and geometric transformations.

Role of the Sulba Sūtras:

The Sulba Sutras are the oldest surviving texts of Indian geometry (c. 800–200 BCE). Their roles include:

1. They encode practical geometry for Vedic ritual: priests needed to construct altars of precise shape and area for different sacrifices.
2. They contain the earliest known statement of the Pythagorean theorem, predating Pythagoras.

3. They introduce geometric transformations: squaring a circle, circling a square, and constructing squares equal in area to rectangles or triangles.
4. They record Pythagorean triples and irrational numbers ($\sqrt{2}$), showing awareness of both rational and irrational quantities.
5. They form the foundation of classical Indian geometry, influencing later mathematical developments by Aryabhata, Brahmagupta, and Bhaskara II.

Q5. Geometric Proof of Baudhayana's Approximation $\sqrt{2} \approx 1 + 1/3 + 1/(3\cdot4) - 1/(3\cdot4\cdot34)$

Baudhayana gave the approximation in the Sulba Sutra (1.61–2): 'The measure is to be increased by a third, and this (third) by its own fourth, less the thirty-fourth part of that fourth.'

$$\sqrt{2} \approx 1 + 1/3 + 1/(3\times4) - 1/(3\times4\times34)$$

$$= 1 + 1/3 + 1/12 - 1/408$$

$$= 408/408 + 136/408 + 34/408 - 1/408$$

$$= 577/408 \approx 1.41421568...$$

Modern value: $\sqrt{2} = 1.41421356...$, so the error is only about 1.5×10^{-7} — remarkable accuracy.

Geometric Construction of $\sqrt{2}$ (Diagonal of a Unit Square):

Consider a unit square ABCD with side = 1. The diagonal AC has length $\sqrt{2}$ by the Pythagorean theorem: $AC^2 = AB^2 + BC^2 = 1 + 1 = 2$.

Baudhayana's Geometric Derivation (Step-by-Step):

6. Start with a unit square of side 1. Divide one side into 3 equal parts, each of length 1/3.
7. The initial approximation to the diagonal of a unit square: 1 whole + 1/3 of 1 = 4/3 (from extending the side by a third).
8. Increase this by a fourth of 1/3: $4/3 + 1/12 = 16/12 + 1/12 = 17/12$.
9. Subtract 1/34 of the last addition (1/12): $17/12 - 1/(12\times34) = 17/12 - 1/408 = 578/408 - 1/408 = 577/408$.

$$577/408 = 1.41421568... \approx \sqrt{2}$$

Verification: $(577)^2 = 332929$ and $(408)^2 \times 2 = 332928$. These differ by only 1, confirming the extraordinary accuracy of this early approximation.

Q6. Construction of a Circle from a Square According to Baudhayana

The problem: Given a square of side a, construct a circle of equal area.

Baudhayana's Rule (Sulba Sutra 2.9):

'If you wish to turn a square into a circle, (take) half its diagonal for the radius; bring it towards the east from the center, and then connect [the arc].' More precisely: the radius r of the circle is obtained as:

$$r = (d/2) - (d/2 - a/2) / 3$$

where $d = a\sqrt{2}$ is the diagonal of the square. Simplifying:

$$r = d/2 - (d - a) / 6 = a\sqrt{2}/2 - a(\sqrt{2} - 1)/6$$

$$r = a[3\sqrt{2} - (\sqrt{2} - 1)] / 6 = a[2\sqrt{2} + 1] / 6$$

Step-by-Step Construction:

10. Draw the square ABCD with side a. Mark its centre O.
11. Draw the diagonal AC. Its half-length (radius of circumscribed circle) = $d/2 = a\sqrt{2}/2$.
12. Let E be the midpoint of one side (e.g., AB), so OE = a/2 (the in-radius).
13. The excess of the circumradius over the in-radius = $d/2 - a/2 = a(\sqrt{2} - 1)/2$.
14. Take one third of this excess: $a(\sqrt{2} - 1)/6$.
15. The required radius = in-radius + one third of excess = $a/2 + a(\sqrt{2} - 1)/6 = a(3 + \sqrt{2} - 1)/6 = a(2 + \sqrt{2})/6$.

Numerical check: For a = 1: $r = (2 + \sqrt{2})/6 \approx (2 + 1.4142)/6 \approx 3.4142/6 \approx 0.5690$.

Area of circle = $\pi r^2 \approx 3.1416 \times 0.3238 \approx 1.0168$. The actual square area = 1. The approximation gives area ≈ 1.017 (about 1.7% error), quite good for the ancient era.

Q7. Construction of a Square from a Rectangle

The problem: Given a rectangle with sides a and b ($a > b$), construct a square with equal area.
The required square has side = $\sqrt{ab}$ (the geometric mean of a and b).
Baudhayana's Method (Gnomon / L-shaped piece method):

16. Draw rectangle ABCD with AB = a (length) and AD = b (breadth).
17. Mark point E on AB such that AE = b. Now square AEFD (side b) is within the rectangle.
18. The remaining strip EBCF has dimensions $(a - b) \times b$.
19. Bisect this strip along its length: cut it into two rectangles each of size $(a - b)/2 \times b$.
20. Place one half-strip along the bottom of the square AEFD (rotated 90°), and the other on the right side.
21. This creates an 'L'-shaped (gnomon) figure around a small square of side $(a - b)/2$.
22. Fill in the corner with a small square of side $(a - b)/2$. The resulting large square has side: $b + (a - b)/2 = (a + b)/2$.
23. Area of large square = $((a+b)/2)^2$. Area of small square corner = $((a-b)/2)^2$. Difference = ab. This difference equals the original rectangle area. The required square of side $\sqrt{ab}$ is obtained by the relation:

$$(\text{Side of required square})^2 = ((a+b)/2)^2 - ((a-b)/2)^2 = ab$$

Geometric realisation: Draw a semicircle on the long side a. Erect a perpendicular from the point b along the diameter. The perpendicular height equals $\sqrt{ab}$ — the side of the required square.

Q8. Sources Used to Reconstruct Indus Valley Mathematics

- Archaeological Excavations: Ruins of cities (Harappa, Mohenjo-daro, Dholavira, Lothal) provide physical evidence of city planning, street grids, and drainage — all implying geometric knowledge.
- Physical Artefacts: Standardised bricks, graduated scales/rulers (Mohenjo-daro scale, Lothal ruler), cubical weights, seals with geometric motifs, and pottery with mathematical patterns.
- Weights and Measures: Discovered cubical and polyhedral weights following a binary-decimal scheme, allowing inference about number systems and metrological practices.
- Architectural Analysis: Measurements of buildings, wells, city blocks, and walls reveal consistent proportional relationships and right-angle construction.
- Comparative Linguistics: The Dravidian language root for 'count' is the same as the stem for the numeral 'eight', suggesting base-8 elements in the proto-Indus counting system.
- Comparison with Contemporary Civilisations: Relationships between Indus, Sumerian, and Mesopotamian units (e.g., the Indus inch = 2 × Sumerian 'sushi') help corroborate metrological reconstructions.
- Later Vedic Texts: Some scholars argue that certain mathematical ideas in the Sulba Sutras preserve earlier Harappan geometric knowledge (e.g., wedge-shaped bricks for circular structures reappear in Vedic altar construction).

Q9. Geometric Features Visible in Indus Town Planning

- Grid Layout: Streets were oriented along cardinal directions (east–west), intersecting at right angles to form a precise grid — the earliest known urban grid planning.
- Standardised Block Sizes: City blocks had uniform dimensions, implying use of a standard unit of length and systematic measurement.
- Citadel–Lower Town Division: Each city was divided into an elevated fortified citadel (upper town) and a lower town — a spatial organisation requiring precise levelling and surveying.
- Right-angle Construction: Building corners and street intersections reveal accurate right angles, likely achieved using ropes (3-4-5 triangle or similar) rather than Pythagorean theorem explicitly.
- Drainage System: Underground covered drains followed precise slopes for effective water flow, demonstrating applied knowledge of gradient and hydraulics.
- Consistent Brick Proportions: All bricks across sites follow the 4:2:1 ratio, requiring standardised molds and understanding of volume and proportion.
- Granary Dimensions: The Mohenjo-daro granary (150 × 75 × 15 feet) shows deliberate 2:1 ratio in plan — mathematically informed architectural planning.

Q10. Two Examples of Geometry from the Indus Valley

Example 1 — Circular Wells with Wedge-shaped Bricks:
At Harappa, circular water wells were constructed using wedge-shaped bricks with isosceles trapezoidal cross-sections. A documented well had an internal diameter of 1.2 m and used 36 bricks, each of length 26 cm covering the circumference. This shows:

$$\textit{Circumference} \approx 36 \times 26\ cm = 936\ cm, \text{ while } \pi \times 120\ cm \approx 376.8\ cm$$

(Note: 36 bricks each spanning approximately 10.5 cm of arc.) This implies knowledge of the relationship between diameter and circumference, i.e., a practical value of $\pi \approx 3.1$.
Example 2 — Space-filling Tiling Patterns on Seals:
A seal from Mohenjo-daro (M-1261) displays a Japanese-fan space-filling tiling pattern. Mathematical analysis shows that the pattern is generated by taking two concentric circles packed in a rectangle of dimensions $4R \times 2R$ and performing a rearrangement of the non-circular regions. The repeating fan-shaped motif has area $2R^2$, and four such motifs rotated by 90° tile the plane. This demonstrates knowledge of:

- Area equivalence (regions cut and rearranged preserve area)
- Rotational symmetry and tiling
- Geometric construction using circle packing

Q11. Did Indus Mathematics Influence Later India?

This is a debated question among historians of mathematics. The evidence is suggestive but not conclusive:
Arguments For Influence:

- Continuity of Brick Technology: Wedge-shaped bricks for circular structures in the IVC reappear in Vedic circular altar construction (described in the Sulba Sutras), suggesting transmission of geometric craft knowledge.
- Weights and Measures: The binary-decimal weight system of the IVC corresponds to weight ratios used in Indian commerce for centuries, persisting into modern times.
- Linguistic Evidence: Dravidian languages (considered descendants of proto-Indus language) retain traces of a base-8 counting system, connecting to Indus numerical culture.
- Geometric Motifs: Star and cross patterns found in Indus seals appear in later Indian decorative traditions.

Arguments Against (or for Caution):

- Chronological Gap: There is a significant gap (c. 1300–1000 BCE) between the late IVC and mature Vedic mathematical texts, making direct transmission uncertain.
- No Textual Link: The Indus script remains undeciphered, so no direct textual transmission can be established.
- Parallel Development: Some geometric similarities (right angles, circle constructions) may represent independent parallel development rather than borrowing.

Conclusion: While direct proof of transmission is absent, the circumstantial evidence — especially in measurement systems, brick geometry, and craft traditions — strongly suggests that Indus mathematical knowledge was not entirely lost but was absorbed into later Vedic and classical Indian culture through craft and oral traditions.

Q12. Decimal-Binary System in Indus Valley Civilisation

The Harappan weight system exhibits a remarkable binary–decimal hybrid structure:
Phase 1 — Binary Progression (doubling):

- 1, 2, 4, 8, 16, 32, 64 units

Each weight is double the previous, forming a geometric series with ratio 2. This allowed any quantity from 1 to 127 units to be measured using at most 7 weights.
Phase 2 — Decimal Progression (continuing from 64):

- 160, 200, 320, 640, 1600, 3200, 6400, 8000, 12800 units

At 160 ($= 16 \times 10$), the sequence transitions to decimal multiples:

- $160 = 16 \times 10$
- $200 = 20 \times 10$
- $320 = 32 \times 10$
- $640 = 64 \times 10$
- $1600 = 64 \times 25$ ($= 160 \times 10$)
- $6400 = 640 \times 10$
- $12800 = 1280 \times 10$

Significance:

- This binary–decimal hybrid anticipated the later Indian decimal system while embedding binary efficiency for small quantities.
- Both 8 and 10 had special counting significance — base-8 traces appear in proto-Dravidian linguistics, while base-10 dominates the large weight classes.
- The unit of weight ≈ 28 grams, close to the English ounce, suggesting possible trade-based metrological convergence.
- Indian currency until the 1950s was also binary (rupee divided into 2, 4, 8, 16, 64 parts), showing the long survival of the Harappan counting tradition.

Q13. Katapayadi System — Identifying Numbers

The Katapayadi System assigns digits 1–9 and 0 to Sanskrit consonants as follows:

Digit	ka-group	ta-group	pa / ya-group
1	ka	ṭa (Ta)	pa, ya
2	kha	ṭha (Tha)	pha, ra
3	ga	ḍa (Da)	ba, la
4	gha	ḍha (Dha)	bha, va
5	ṅa (nga)	ṇa (Na)	ma, śa
6	ca (cha)	ta	ṣa, (Sha)
7	cha (Cha)	tha	sa
8	ja	da	ha
9	jha	dha	—
0	ña (nya)	na	—

Rules: (1) Only consonants carry digit values; vowels alone = 0. (2) In a conjunct consonant cluster, only the last consonant counts. (3) The number is read RIGHT TO LEFT (ankanam vamato gatih — 'digits go leftward').

Decoded Values (Consonants → Digits → Reversed):

a) Līlāvatī: l→3, l→3, v→4, t(i)→6
Digits: 3,3,4,6 → Reversed: 6,4,3,3 → **Number = 6433**

b) Anuyogadvārasūtra: n→0, y→1, g→3, d→8, r→2, s→7, t→6
Digits: 0,1,3,8,2,7,6 → Reversed: 6,7,2,8,3,1,0 → **Number = 6728310**

c) Sulbasutras: s→7, l→3, b→3, s→7, t→6, r→2, s→7
Digits: 7,3,3,7,6,2,7 → Reversed: 7,2,6,7,3,3,7 → **Number = 7267337**

d) Brāhmasphuṭasiddhānta: r→2, h→8, m→5, s→7, p→1, h→8, ṭ→1, s→7, d→8, dh→9, n→0, t→6
Digits: 2,8,5,7,1,8,1,7,8,9,0,6 → Reversed: 6,0,9,8,7,1,8,1,7,5,8,2 → **Number = 609871817582**

e) Anandamrtam: n→0, n→0, d→8, m→5, r→2, t→6, m→5
Digits: 0,0,8,5,2,6,5 → Reversed: 5,6,2,5,8,0,0 → **Number = 5625800**

f) Rakshabandhanotsavah: r→2, k→1, sh→5, b→3, n→0, dh→9, n→0, ts→7, v→4, h→8
Digits: 2,1,5,3,0,9,0,7,4,8 → Reversed: 8,4,7,0,9,0,3,5,1,2 → **Number = 8470903512**

g) Korona Rajanaskah: k→1, r→2, n→0, r→2, j→8, n→0, s→7, k→1, h→8
Digits: 1,2,0,2,8,0,7,1,8 → Reversed: 8,1,7,0,8,2,0,2,1 → **Number = 817082021**

h) Aryabhatiya: r→2, y→1, bh→4, ṭ→1, y→1

Digits: 2,1,4,1,1 → Reversed: 1,1,4,1,2 → **Number = 11412**

i) Chandahasastra: ch→6, n→0, d→8, h→8, s→7, s→7, t→6, r→2
Digits: 6,0,8,8,7,7,6,2 → Reversed: 2,6,7,7,8,8,0,6 → **Number = 26778806**

j) Ganita-Sara-Sangraha: g→3, n→0, t→6, s→7, r→2, s→7, n→0, g→3, r→2, h→8
Digits: 3,0,6,7,2,7,0,3,2,8 → Reversed: 8,2,3,0,7,2,7,6,0,3 → **Number = 8230727603**

k) Siddhanta Shiromani: s→7, d→8, dh→9, n→0, t→6, sh→5, r→2, m→5, n→0
Digits: 7,8,9,0,6,5,2,5,0 → Reversed: 0,5,2,5,6,0,9,8,7 → **Number = 052560987**

l) Suryaprajnapti: s→7, r→2, y→1, p→1, r→2, j→8, n→0, p→1, t→6
Digits: 7,2,1,1,2,8,0,1,6 → Reversed: 6,1,0,8,2,1,1,2,7 → **Number = 610821127**

Q14. Bhūta-Saṃkhyā System — Identifying Numbers

In the Bhūta-Saṃkhyā (object-numeral) system, familiar objects, nature, and philosophy are used as mnemonic number symbols. Like the Katapayadi system, digits are also read right to left (ankanam vamato gatih).

Standard word–digit mappings used below (from Table 6.3):

- 0: śūnya, anata, pūrna, kha
- 1: ādi, candra, prithivi, eka
- 2: aśvin, ayana, dvandva, dvi
- 3: rāma, guṇa, loka, kāla, agni, trinetra
- 4: veda, śruti, yuga, āśrama, varṇa, samudra, kṛta
- 5: bhūta, śāstra, bāṇa, pāṇḍava, indriya
- 6: aṅga, ṛtu, darśana, ṣaṇmukha, ṣaṭ
- 7: ṛṣi, adri, svara, dhātu, chandas
- 8: vasu, bhujaṅga, siddhi, dik, kuñjara, nāga
- 9: gṛha, aṅka, nanda

a) rama-chandra-guna-nanda-risi-vedah:
rāma=3, candra=1, guṇa=3, nanda=9, ṛṣi=7, veda=4
Digits in sequence: 3, 1, 3, 9, 7, 4 → Read right to left: 4, 7, 9, 3, 1, 3

Number = 479313

In Katapayadi: 4→gha/dha, 7→sa, 9→jha/dha, 3→ga/ba, 1→ka/ya, 3→ga/ba

Possible Katapayadi encoding: dha-sa-jha-ga-ka-ga (e.g., 'dhasa-jhagaka-ga')

b) guna-adri-kha-sastra-nayana-bhujangah:
guṇa=3, adri=7, kha=0, śāstra=5, nayana=2, bhujaṅga=8
Digits: 3, 7, 0, 5, 2, 8 → Right to left: 8, 2, 5, 0, 7, 3

Number = 825073

Katapayadi encoding: ha-dvi-panca-sunya-risi-ga → 'ha-ra-sha-na-ra-ga' (approximate)

c) purna-tri-veda-gruha-nanda-rtu-padah:
pūrna=0, tri=3, veda=4, gṛha=9, nanda=9, ṛtu=6, pada=1
Digits: 0, 3, 4, 9, 9, 6, 1 → Right to left: 1, 6, 9, 9, 4, 3, 0

Number = 1699430

Katapayadi encoding: ya-sha-nanda-nanda-veda-ga-na → 'ya-ṣa-dha-dha-va-ga-na' (approximate)

Q15. Numbers Rewritten in Bhūta-Saṃkhyā System

Method: Write each digit (right to left in the number, i.e., units digit first) as a word from Table 6.3.

a) 179,532,695:
Digits (right to left): 5, 9, 6, 2, 3, 5, 9, 7, 1

- 5 → bāṇa (arrow / five arrows of Kāmadeva)
- 9 → nanda

- 6 → ṛtu (seasons)
- 2 → aśvin (twin gods)
- 3 → rāma
- 5 → indriya
- 9 → grha (planets)
- 7 → ṛṣi (sages)
- 1 → ādi (beginning)

bana-nanda-rtu-asvin-rama-indriya-grha-risi-adi

b) 345,447:
Digits (right to left): 7, 4, 4, 5, 4, 3

- 7 → svara (musical notes)
- 4 → veda
- 4 → yuga
- 5 → bhūta (five elements)
- 4 → āśrama
- 3 → guṇa (three qualities)

svara-veda-yuga-bhuta-asrama-guna

c) 670,087,123:
Digits (right to left): 3, 2, 1, 7, 8, 0, 0, 7, 6

- 3 → trinetra
- 2 → dvi
- 1 → eka
- 7 → chandas (metres)
- 8 → vasu
- 0 → śūnya
- 0 → kha
- 7 → adri (mountains)
- 6 → aṅga (limbs of Veda)

trinetra-dvi-eka-chandas-vasu-sunya-kha-adri-anga

Q16. Speed of Light from Sayanacharya's Commentary

Sayanacharya (14th century CE) wrote in his commentary on the Rigveda: 'In half a nimesa the sun traverses 2202 yojanas.'

Step 1 — Duration of half a nimesa:
From the ancient Indian time unit chain (Chapter 3):

- 1 Paramanu = 1.3133×10^{-5} s
- 3 Vedha = 1 Lava; 3 Lava = 1 Nimesa
- 1 Nimesa ≈ 0.2127 s (approximately 16/75 s)
- Half a Nimesa ≈ 0.1064 s

Step 2 — Distance covered:
1 Yojana = 14.484 km (from Chanakya's Arthashastra via Angula-Dhanusha conversion)

$$\text{Distance} = 2202 \text{ yojanas} \times 14.484 \text{ km/yojana} = 31{,}893.5 \text{ km}$$

Step 3 — Speed calculation:

$$\text{Speed} = \text{Distance} / \text{Time} = 31{,}893.5 \text{ km} / 0.1064 \text{ s} \approx 299{,}845 \text{ km/s}$$

Comparison with Modern Value:

Sayanacharya's value	≈ 299,845 km/s
Modern value	299,792 km/s
Percentage error	≈ 0.018% — remarkably close!

The accuracy is remarkable for a 14th-century text. The slight deviation arises from the specific values used for the Yojana and Nimesa.

Q17. Vedic Sutras — Solved Problems

a) Paraavartya Yojayet: 36520981 ÷ 133

'Paraavartya Yojayet' means 'Transpose and Apply'. It is used for division where the divisor does not end in 9. The method involves rewriting the divisor as 100 + 33 and handling carries.

Direct computation:

24. 133 × 200,000 = 26,600,000
25. 36,520,981 − 26,600,000 = 9,920,981
26. 133 × 70,000 = 9,310,000 → 9,920,981 − 9,310,000 = 610,981
27. 133 × 4,000 = 532,000 → 610,981 − 532,000 = 78,981
28. 133 × 500 = 66,500 → 78,981 − 66,500 = 12,481
29. 133 × 90 = 11,970 → 12,481 − 11,970 = 511
30. 133 × 3 = 399 → 511 − 399 = 112

$$36{,}520{,}981 \div 133 = 274{,}593 \text{ remainder } 112$$

Verification: 274,593 × 133 = 36,520,869; 36,520,869 + 112 = 36,520,981 ✓

b) Sopaantyadvayamantyam: 624 × 12

'Sopaantyadvayamantyam' means 'The ultimate and twice the penultimate'. This sutra handles special multiplication forms. Here:

624 × 12 = 624 × (10 + 2) = 6240 + 1248

$$624 \times 12 = 7{,}488$$

Alternatively using Vertically and Crosswise:

6|2|4 × 1|2: (6×1)|(6×2 + 2×1)|(2×2 + 4×1)|(4×2) = 6|14|8|8 → carry: 6|14|8|8 → 7488

c) Yaavadunam: 989 × 1005

'Yaavadunam' means 'By the deficiency'. Express numbers in relation to a base (here base = 1000):

- 989 = 1000 − 11 (deficiency = 11)
- 1005 = 1000 + 5 (surplus = 5)

Apply the sutra: $(a-d_1)(b+d_2) = ab + ad_2 - bd_1 - d_1d_2$

$$989 \times 1005 = (1000-11)(1000+5)$$
$$= 1000\times1000 + 1000\times5 - 11\times1000 - 11\times5$$
$$= 1{,}000{,}000 + 5{,}000 - 11{,}000 - 55$$
$$= 993{,}945$$

Verification: 989 × 1000 = 989,000; 989 × 5 = 4,945; total = 993,945 ✓

d) Dhvajanka: 601325 ÷ 76

'Dhvajanka' (Flag Method) is used when the divisor has multiple digits. Flag digit = 6 (units of 76); working digit = 7 (tens).

Step-by-step division:

31. 76 × 7000 = 532,000 → 601,325 − 532,000 = 69,325
32. 76 × 900 = 68,400 → 69,325 − 68,400 = 925
33. 76 × 12 = 912 → 925 − 912 = 13

$$601{,}325 \div 76 = 7{,}912 \text{ remainder } 13$$

Verification: 7,912 × 76 = 553,840 + 47,472 = 601,312; 601,312 + 13 = 601,325 ✓

Q18. Cyclic Quadrilateral — Brahmagupta's Diagonals, Area, and Circumradius

Given: Sides a = 39, b = 52, c = 60, d = 25 of a cyclic quadrilateral.

Step 1 — Compute Products:

Product	Expression	Value

ab	39 × 52	**2028**
cd	60 × 25	**1500**
ac	39 × 60	**2340**
bd	52 × 25	**1300**
ad	39 × 25	**975**
bc	52 × 60	**3120**
ab + cd		**3528**
ac + bd		**3640**
ad + bc		**4095**

Step 2 — Diagonals (Brahmagupta's Formula):

$$p^2 = (ab+cd)(ac+bd) / (ad+bc) = 3528 \times 3640 / 4095 = 12,841,920 / 4095 = 3136$$

$$p = \sqrt{3136} = 56$$

$$q^2 = (ad+bc)(ac+bd) / (ab+cd) = 4095 \times 3640 / 3528 = 14,905,800 / 3528 = 4225$$

$$q = \sqrt{4225} = 65$$

Step 3 — Area (Brahmagupta's Area Formula):

$$s = (a + b + c + d) / 2 = (39 + 52 + 60 + 25) / 2 = 176 / 2 = 88$$

$$Area\ K = \sqrt{[(s-a)(s-b)(s-c)(s-d)]}$$

$$= \sqrt{[(88-39)(88-52)(88-60)(88-25)]}$$

$$= \sqrt{[49 \times 36 \times 28 \times 63]}$$

$$= \sqrt{[49 \times 36 \times 1764]}$$

$$= 7 \times 6 \times 42 = 1764 \text{ square units}$$

Step 4 — Circumradius:

$$R = \sqrt{[(ab+cd)(ac+bd)(ad+bc)]} / (4K)$$

$$Numerator = \sqrt{[3528 \times 3640 \times 4095]}$$

$$= \sqrt{[3528 \times 3640 \times 4095]} = \sqrt{[52,587,662,400]}$$

Factoring: $3528 = 8\times441 = 8\times21^2$, $3640 = 8\times455 = 8\times5\times7\times13$, $4095 = 5\times9\times7\times13$

$$= \sqrt{[64 \times 21^2 \times 25 \times 49 \times 9 \times 169]} = 8 \times 21 \times 5 \times 7 \times 3 \times 13 = 229,320$$

$$R = 229,320 / (4 \times 1764) = 229,320 / 7,056 = 32.5 \text{ units}$$

Summary of Results:

Diagonal p	**56 units**
Diagonal q	**65 units**
Area K	**1764 sq. units**
Circumradius R	**32.5 units**